Computer Programming Lab Report

Course: EE-170L Computer Programming Lab

Lab Title: Variables and Basic C++ Programs

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Date: ____________________

1. Objective

This lab aims to understand variable declaration, memory size, memory location, floating-point variables, and temperature conversion using C++ programming language.

2. Theory

A variable is a named memory location used to store data. Each variable has a data type which determines the type of value it can store and the amount of memory it occupies. Common data types include int, float, double, char, and bool. The sizeof() operator is used to determine memory size, and the address operator (&) is used to find memory location.

3. Software Used

Visual Studio Code, C++ Compiler (G++), and Windows Operating System.

4. Procedure

Open Visual Studio Code, create a .cpp file, write the program, compile using compiler, run the program, and observe the output.

5. Task 1: Variable Declaration, Size, Location and Value

```
#include <iostream>
using namespace std;

int main()
{
    int a = 10;
    float b = 5.5;
    double c = 20.99;
    char d = 'A';
    bool e = true;

    cout << "Integer value: " << a << endl;
    cout << "Size: " << sizeof(a) << endl;
    cout << "Location: " << &a << endl;

    cout << "Float value: " << b << endl;
    cout << "Size: " << sizeof(b) << endl;
    cout << "Location: " << &b << endl;

    return 0;
}
```

6. Task 2: Floating Point Variable PI

```
#include <iostream>
using namespace std;
int main()
```

```
{
    float pi = 3.14;
    cout << "Value of pi: " << pi << endl;
    return 0;
}
```

7. Task 3: Celsius to Fahrenheit Conversion

```
#include <iostream>
using namespace std;

int main()
{
    float celsius, fahrenheit;
    cout << "Enter Celsius: ";
    cin >> celsius;
    fahrenheit = (celsius * 9/5) + 32;
    cout << "Fahrenheit: " << fahrenheit << endl;
    return 0;
}
```

8. Conclusion

This lab successfully demonstrated how to declare variables, determine their memory size and location, use floating-point variables, and perform temperature conversion using C++.